

Full Stack Developer Task – PHP / Laravel (Fresher)

Project: Simple Chat Application (WhatsApp Web Inspired)

Project Overview

Build a simplified chat application inspired by WhatsApp Web that allows two users to send and receive messages. The objective of this assignment is to evaluate your understanding of PHP, Laravel fundamentals, database handling, and basic frontend integration. You may use AI tools while building the project.

Technical Requirements

1. User Setup

Implement basic user handling:

- Create users using a simple registration or login mechanism
- Each user should have a unique identifier
- Support chat between at least two users
- Users should be identifiable during conversations

Authentication can be implemented using **Laravel authentication or a simple login system**.

2. Chat Interface

Create a simple chat interface:

- Display list of available users or chats
- Select a user to start a conversation
- Chat window showing message history
- Input box to send messages
- Sent and received messages should appear differently
- Messages should remain visible after page refresh

3. Messaging Functionality

Implement the following features:

- Send text messages between users
- Store messages in the database
- Fetch messages for the selected chat
- Display messages in chronological order

Each message should contain:

1. Sender ID
2. Receiver ID
3. Message text

4. Timestamp

4. Backend Implementation

Use Laravel to build the backend:

- Create **Models, Controllers, and Migrations**
- Use **Laravel routes and controllers** to handle chat actions
- Validate message inputs
- Handle empty or invalid requests properly

5. Application Structure

Maintain a clean project structure:

- Follow **Laravel MVC architecture**
- Organize controllers and models properly
- Use migrations for database tables
- Ensure readable and structured code

Technology Stack

Frontend

- Blade templates / basic HTML + CSS
- JavaScript (optional for better UI)

Backend

- PHP
- Laravel Framework
- MySQL

Submission Instructions

- Submit the completed project as a **public GitHub repository**.
- Your repository should include:
 1. Source code
 2. Database migration files
 3. A README file explaining:
 - How to run the project locally
 - Database setup instructions
 - Environment configuration